Logbook for week beginning 19/10/2026

20/10

Meeting with Dr. Seeburger and Dr. Kamann.

Notes:

- Identified a Be type star as potential cause behind Hα double peak in 4295254307232999296. This is due to the rapid rotation of the star causing a decretion disk.
- Discussed potential cause behind inconsistent absorption between observations for 6609971199873698944.
- Discussed parameters needed for the future: metallicity, temperature and log(g).
- Discussed methods for cross-correlation that continuum needs to be subtracted, template must be normalised in the same method as observed spectrum and the use of SciPy package for Fast Fourier Transform (FFT). Recommended Tonry & Davis, 1979 as the method.
- Discussed method of measuring radial velocity once cross-correlation has been completed.

To obtain an estimate for metallicity, temperature and log(g) for each of the targets in UVES, I queried the Gaia database for `gspspec` and `gspphot`.

Gaia ID	T_{spec} (K)	$T_{phot}(K)$	$[M/H]_{spec}$	$[M/H]_{phot}$	$\log(g)_{spec}$	$\log(g)_{phot}$
4295254307232999296	–	12395	–	−0.601	–	3.729
4039283116865495424	–	6533	–	−0.102	–	4.008
4239611700214875904	–	–	–	–	–	–
4973506334166805376	6450	–	–	–	4.300	–
6551893071549223040	–	–	–	–	–	–
6609971199873698944	–	6059	–	−0.247	–	4.085

This tells me that when looking for synthetic spectra, I am looking for one where $T \approx 6000 - -7000\,K$ and $[M/H] \approx -0.1 - -0.3$, and another where $T \approx 12000\,K$ and $[M/H] \approx -0.6$, with both having log(g) ≈ 3.5 – 4.5. The first database I looked at was the PHOENIX catalogue (arXiv:1303.5632).

Choosing `lte06500-4.00-0.0.PHOENIX-ACES-AGSS-COND-2011-HiRes` with $T_{eff} = 6500K$, $\log(g) = 4.0$ and $[M/H] = 0.0$. Cutting the synthetic it into the desired bands of $4700\,\text{Å} < \lambda < 6900\,\text{Å}$ to match the UVES wave-range and cutting at 5760 Å to mimic the chip gap. Normalising this using the method the observed spectra used gives

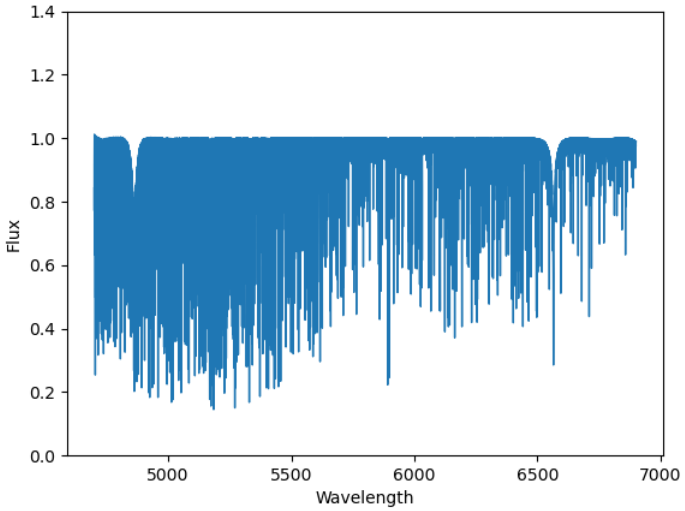


Figure 1: Normalised PHOENIX synthetic spectra.

Another synthetic spectra I tried was from the BOSZ catalogue (DOI: 10.17909/T95G68), specifically `bosz_t6500_g4.0_m0.00`, where $T_{eff} = 6500K$, $\log(g) = 4.0$ and $[M/H] = 0.0$.

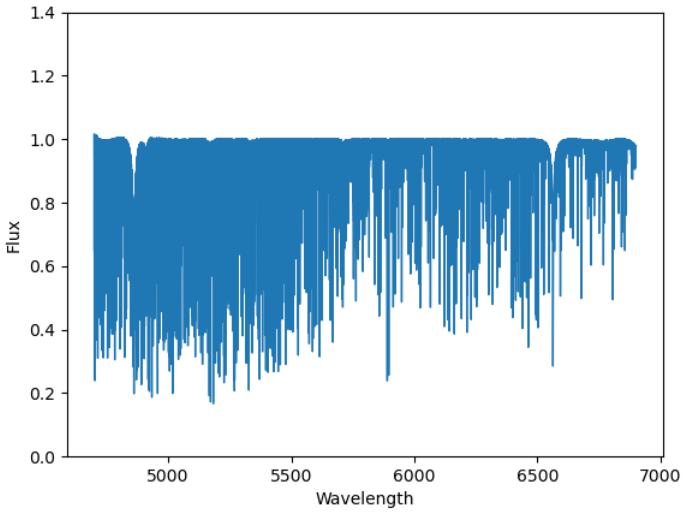


Figure 2: Normalised BOSZ synthetic spectra.

Both of these spectra were degraded using `gaussian_filter` and `interp1d` from `scipy`. These spectra were put into log space and degraded to $R = 66,320$ to match the resolution of each UVES spectrum.

21/10

Fixed issue with `cosmic_rays` as it only identified pixels with flux $> 2\,\sigma$ of median and didn't remove them. It now finds any with flux $> 5\,\sigma$ of continuum and removes them.

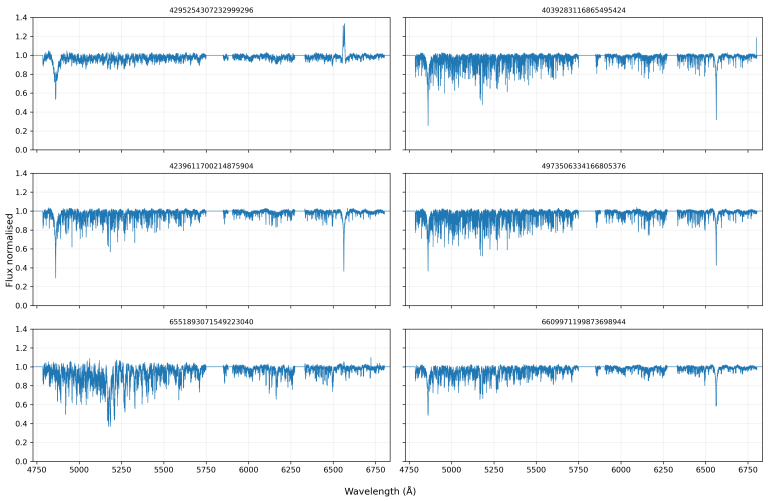


Figure 3: Normalised spectra grid. Na D, Telluric O_2 line and chip gap all removed.

22/10

Updated `degrade_spectra` function to include the resolution of the synthetic spectra for more accurate degradation.

Weekly summary

Successfully degraded some synthetic spectra to the resolution of the UVES data. I succeeded in completing the task set out for me during the meeting, and resultantly failed to complete the task of satisfying assumption 6 I set last week.

Files generated: Synthetic_test.py, degraded spectra of BOSZ and PHOENIX synthetic spectra, full wavelength range spectra of all candidates.

Synthetic_test.py required modules: numpy, pandas, matplotlib, astropy, scipy.

References: arXiv:1303.5632, DOI: 10.17909/T95G68.

Corrections from previous week(s)

Corrected `cosmic_rays` function so that it correctly implemented a sigma-cut filter above continuum. Corrected `line_marking` function to also remove telluric O_2 lines and the chip gap.

Tier 2 AI Policy:

I used Claude AI for bug fixing and planning code. I critically evaluated all outputs and remain responsible for the submitted work.